Evaluating Cross-Lingual Performance and Implicit Translation in Large Language Models

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Abstract

Large language models (LLMs) are trained predominantly on English data yet deployed in multilingual contexts, raising open questions about how well they generalize across languages and whether they rely on implicit translation mechanisms when processing non-English inputs. We study these questions through a two-pronged framework: black-box accuracy evaluation and white-box mechanistic probing via logit lens and Representational Similarity Analysis (RSA). We construct a 180-question benchmark spanning three languages—English, Spanish, and Basque—and three task categories (math, factual, reasoning), and evaluate two 7B-parameter instruction-tuned models: Mistral-7B-Instruct-v0.2 and Qwen2-7B-Instruct. Mistral achieves 68% overall English accuracy but collapses to 8% on Basque (0% factual); Qwen2 is significantly more robust at 37% Basque overall. Logit lens reveals that math answers are identified at shallow layers regardless of language, while Basque factual representations become structurally random by the final layer. RSA across all 32 layers provides evidence for implicit translation in reasoning tasks: English-Spanish representational similarity peaks in mid-layers (Spearman $r = 0.81$ at layer 8) before declining at the output stage, consistent with a shared cross-lingual computation resolved in a language-specific way. Together, our results show that cross-lingual performance disparities are not merely behavioral but are reflected deep in the residual stream, and that the degree of cross-lingual representational alignment differs substantially by task category and language pair.

1 Introduction

The rapid deployment of large language models (LLMs) in multilingual settings has outpaced our understanding of how these models process non-English content [Ahuja et al., 2023, Huang et al., 2023]. While models such as Mistral-7B and Qwen2-7B are widely used across dozens of languages, they are trained on corpora that are heavily weighted toward English. This creates an empirical gap: we know behavioral accuracy often drops for non-English inputs, but the mechanistic story—*why* and *how* does processing fail, and at what layer?—remains largely unexplored.

A central hypothesis in the field is that instruction-tuned multilingual LLMs engage in *implicit internal translation*: when given a non-English input, the model maps it to an internal English-like representation before computing an answer, then (in theory) projects back to the target language for output [Wendler et al., 2024, Tang et al., 2024]. This hypothesis has important implications for fairness and reliability—if a model cannot construct the intermediate representation for a low-resource language like Basque, it may tokenize the input successfully yet fail to process it meaningfully.

Our work investigates two complementary questions:

1. **Behavioral:** How do accuracy and generation quality vary across English, Spanish, and Basque, and across math, factual, and reasoning task categories?
2. **Mechanistic:** Do the residual stream representations of semantically equivalent questions across languages share

relational structure (RSA), and does cross-lingual representational similarity arise and decay at predictable layers (logit lens)?

We construct a 180-question benchmark of aligned triplets (English / Spanish / Basque), run two 7B models on GPU via Modal with full hidden-state extraction at every layer, and present a joint behavioral + mechanistic analysis. Our key contributions are:

1. A lightweight multilingual benchmark covering three languages and three task categories, with LLM-as-judge evaluation (GPT-4o-mini / Grok-3-mini).
2. Logit lens trajectories showing that math answers emerge at shallow layers independent of language, while Basque factual representations fail to converge even by the final layer.
3. RSA analysis providing evidence for implicit translation in reasoning tasks: English-Spanish representations converge in middle layers before diverging at output.
4. A cross-model comparison showing that Mistral and Qwen2 develop qualitatively different internal representations for multilingual content, even when overall accuracy is similar.

2 Related Work

Multilingual LLM benchmarks. Most established benchmarks for LLM evaluation are English-centric, with multilingual variants often constructed via translation rather than native composition [Hendrycks et al., 2020]. Recent efforts such as INCLUDE [Romanou et al., 2024] and MMLU-ProX [Chen et al., 2025] evaluate LLMs across dozens of languages but focus on behavioral accuracy rather than internal representations. Our benchmark is smaller but paired with white-box probing, allowing mechanistic interpretation of behavioral gaps.

Cross-lingual generalization. Ahuja et al. [2023] conduct a large-scale multilingual evaluation across 16 models and 70 languages, finding consistent degradation for low-resource languages. Kargaran et al. [2024] introduce Mexa, measuring alignment between English and non-English representations in pre-trained models. We build on this direction by analyzing alignment at every layer of the residual stream and connecting it to task-specific behavioral accuracy.

Implicit translation hypothesis. Wendler et al. [2024] provide evidence that LLaMA models internally represent non-English tokens via English-centric latent spaces using logit lens probing. Tang et al. [2024] study language-agnostic and language-specific neurons in multilingual transformers. Our RSA-based analysis provides complementary evidence that cross-lingual representational structure peaks at intermediate depths, specifically for reasoning tasks, consistent with a mid-layer implicit translation stage.

Logit lens and residual stream analysis. The logit lens technique [nostalgebraist, 2020] projects intermediate hidden states through the final layer norm and language model head to decode what the model "predicts" at each layer. Din et al. [2023] apply it to study factual association retrieval. We extend this approach to a multilingual setting, comparing ground-truth token rank trajectories across three languages and two models with different vocabulary sizes.

Representational Similarity Analysis. RSA [Kriegeskorte et al., 2008], originally from computational neuroscience, has been applied to compare representational geometry across neural network layers and models [Kornblith et al., 2019]. We use layer-wise RSA to measure how much the *relational structure* among questions is shared across languages—a more nuanced probe than direct vector similarity, since it measures whether the model treats analogous question pairs as analogously (dis)similar.

3 Experimental Design

3.1 Benchmark Construction

We constructed a benchmark of 180 questions organized as aligned triplets: each question appears in English, Spanish, and Basque, sharing the same ground-truth answer. Questions are split into three categories (20 questions each per language):

- **Math:** Arithmetic, algebra, and geometry problems. Answers are numerals or short symbolic expressions, minimizing lexical variation across languages.
- **Factual:** World knowledge questions covering science, geography, and history. Requires retrieval of facts likely represented unevenly across languages in the training corpus.
- **Reasoning:** Logic puzzles, lateral thinking, and trick questions. Requires structural inference; we hypothesize reasoning structure may be shared cross-lingually in intermediate layers.

We selected a three-language gradient to span high-resource (English), medium-resource (Spanish, Indo-European), and low-resource (Basque, language isolate with no close relatives and minimal training data representation). Original questions were written in English and translated to Spanish and Basque via GPT-4, with manual verification to ensure semantic equivalence and avoid translation artifacts.

3.2 Models

We evaluate two 7B-parameter instruction-tuned models:

- **Mistral-7B-Instruct-v0.2:** 32 transformer layers, $d_{\text{model}} = 4096$, vocabulary size 32,000. We use the `[INST]...[/INST]` prompt format.
- **Qwen2-7B-Instruct:** 28 transformer layers, $d_{\text{model}} = 3584$, vocabulary size 152,064 (a significantly larger multilingual vocabulary). We use the Qwen2 chat template.

Both models were run on A100 GPUs via Modal in `float16` with greedy decoding and `max_new_tokens=50`. We use a single unified model for both black-box accuracy and all white-box probing, avoiding the cross-model confound present in our early-phase experiments where we used GPT-3.5-turbo and Gemini for inference but BLOOM-560M for probing.

3.3 Black-Box Evaluation

For each of the 180 questions, we record the model's free-form generation. Answers are evaluated using an LLM-as-judge framework: we submit each (ground truth, model answer) pair to GPT-4o-mini (with Grok-3-mini as fallback) with a binary YES/NO prompt. Results are cached in `*_accuracy_cache.json` to ensure reproducibility. Accuracy is aggregated by language and category.

3.4 Logit Lens

At each of the 32 (Mistral) or 28 (Qwen2) layers, we project the last-token hidden state $\mathbf{h}^{(\ell)}$ through the final layer normalization and the language model head W_{lm}, then apply softmax:

$$p^{(\ell)} = \text{softmax}(W_{\text{lm}} \cdot \text{LayerNorm}(\mathbf{h}^{(\ell)})).$$

We record two metrics per question per layer:

- **Ground-truth rank (GT rank):** The rank of the first ground-truth answer token in $p^{(\ell)}$ (lower is better; rank 1 means the model already "points at" the correct answer).
- **Top-1 confidence:** $\max_t p_t^{(\ell)}$, indicating how decisive the model is at each layer.

3.5 Representational Similarity Analysis (RSA)

For each (language, category) group of 20 questions, we extract the last-token residual stream vector $\mathbf{h}_i^{(\ell)} \in \mathbb{R}^{d_{\text{model}}}$ at each layer ℓ with `output_hidden_states=True`, skipping the embedding layer. We then build a 20×20 Representational Dissimilarity Matrix (RDM):

$$\text{RDM}_{ij}^{(\ell)} = 1 - \cos(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}).$$

For each language pair $(\mathcal{L}_1, \mathcal{L}_2)$ and each category, we compute the Spearman rank correlation between $\text{RDM}_{\mathcal{L}_1}^{(\ell)}$ and $\text{RDM}_{\mathcal{L}_2}^{(\ell)}$ at every layer, yielding a cross-lingual alignment curve. A high correlation indicates that the model organizes the relational structure among questions similarly for both languages.

4 Results

4.1 Black-Box Accuracy

Table 1 and Figure 1 summarize black-box accuracy across all language-category combinations for both models.

Table 1: Black-box accuracy (%) by language, category, and model. **Bold** indicates best result per cell. Qwen2-7B is substantially more robust on Basque across all categories.

Language	Category	Mistral-7B	Qwen2-7B
English	Math	45	**55**
	Factual	**100**	**100**
	Reasoning	**60**	**60**
	Overall	68	**72**
Spanish	Math	50	**70**
	Factual	**100**	95
	Reasoning	35	**65**
	Overall	62	**77**
Basque	Math	20	**65**
	Factual	0	**30**
	Reasoning	5	**15**
	Overall	8	**37**

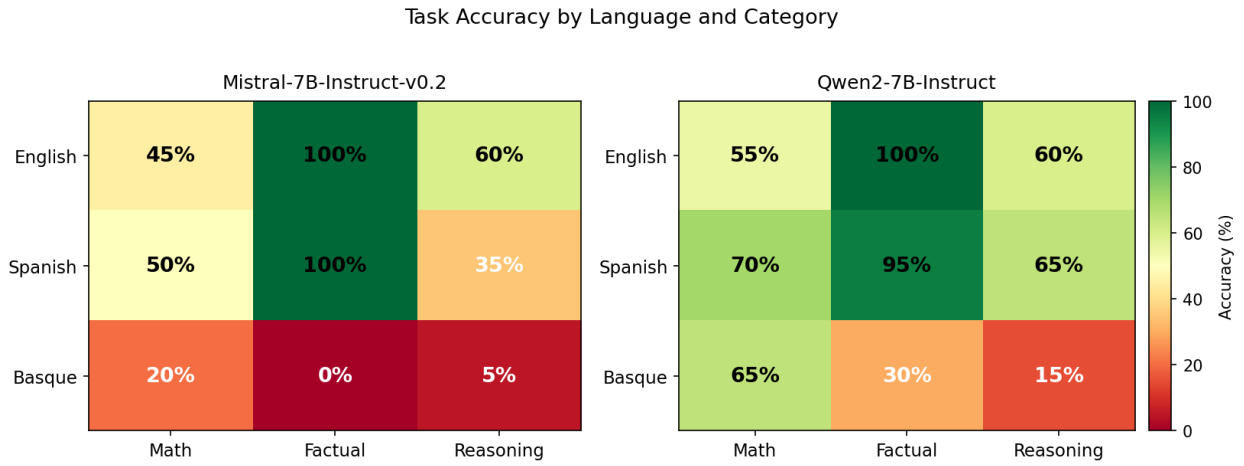

Figure 1: Accuracy heatmaps for Mistral-7B (left) and Qwen2-7B (right). Rows are languages, columns are task categories. Qwen2-7B's advantage is most dramatic on Basque, particularly for math (65% vs. 20%) and factual (30% vs. 0%). Both models achieve 100% English factual accuracy.

Several patterns stand out. First, **Basque causes dramatic collapse for Mistral**: 8% overall accuracy and 0% factual accuracy, compared to 68% and 100% for English. Spanish is intermediate (62% overall), roughly commensurate with its training data representation. Second, **Qwen2 is substantially more multilingual for Basque**: 37% overall (vs. 8% for Mistral), and 65% Basque math (vs. 20%). This likely reflects Qwen2's larger and more multilingual training corpus, though exact composition is not publicly disclosed. Third, **factual knowledge is the most vulnerable category for low-resource languages**: both models achieve 0–30% Basque factual accuracy, whereas Basque math achieves 20–65%. The concrete numerical nature of math problems provides a partial hedge against language-specific knowledge gaps.

4.2 Logit Lens: Layer-by-Layer Prediction Trajectories

Table 2: Logit lens summary for Mistral-7B-Instruct-v0.2: mean ground-truth rank (lower is better) and mean top-1 confidence at layers 1 and 32.

		GT Rank		Top-1 Conf.	
Language	**Category**	**L1**	**L32**	**L1**	**L32**
English	Math	1,160	261	0.001	**0.942**
	Factual	12,041	**26**	0.001	**0.978**
	Reasoning	5,652	186	0.001	0.870
Spanish	Math	1,143	452	0.001	0.810
	Factual	14,912	219	0.001	0.768
	Reasoning	11,241	309	0.001	0.688
Basque	Math	**339**	**62**	0.001	0.703
	Factual	10,434	2,440	0.001	0.848
	Reasoning	5,690	108	0.001	0.539

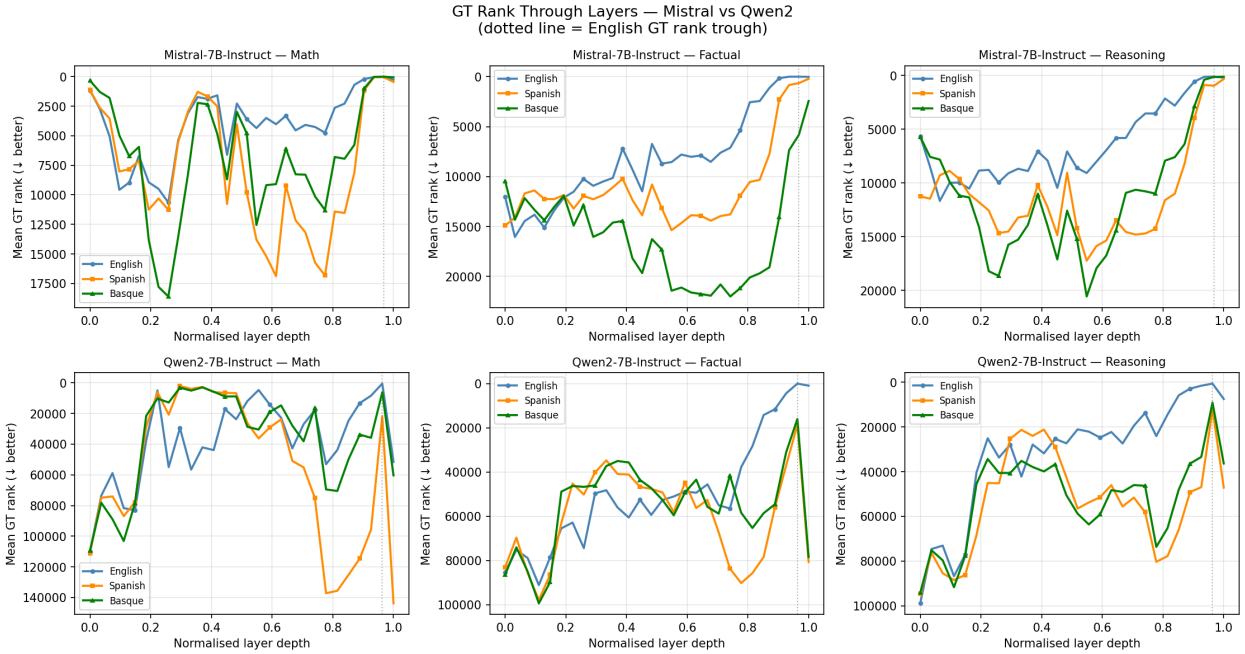

Figure 2: Mean ground-truth token rank as a function of normalized layer depth, for all languages and categories, for Mistral-7B (top row) and Qwen2-7B (bottom row). Lower rank is better. The Basque factual curve remains elevated throughout all layers in Mistral, while Qwen2 shows improved convergence. Note Qwen2's higher absolute ranks reflect its larger vocabulary (152K tokens).

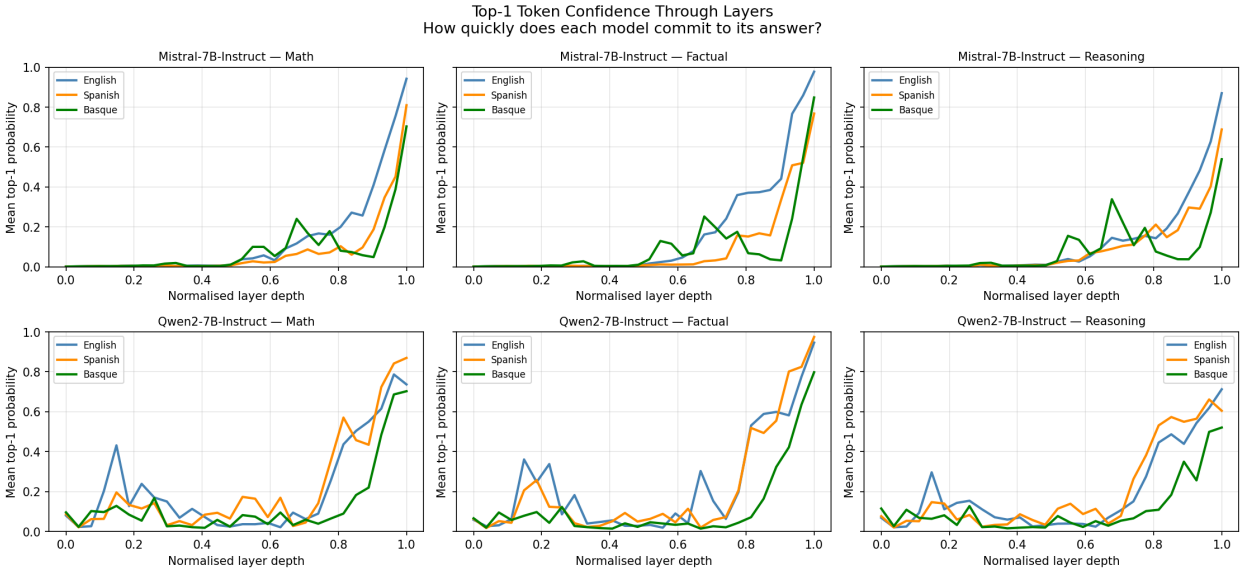

Figure 3: Mean top-1 token probability (confidence) as a function of layer, for both models. All languages converge to near-zero confidence in early layers (uniform distribution over vocabulary), then sharply increase in later layers. Basque reasoning maintains the lowest final-layer confidence for Mistral (53.9%), consistent with its 5% accuracy.

Three key findings emerge from Table 2 and Figures 2 and 3:

Finding 1: Math is language-agnostic in early layers. Basque math has a *lower* mean GT rank at layer 1 (339) than English (1,160) or Spanish (1,143), meaning the model already gestures toward the correct answer earlier in Basque math than in English math. This is counterintuitive given Basque's low accuracy, but it makes sense: math answers are numerals, which tokenize identically regardless of surrounding language. The early residual stream quickly identifies numeric answer tokens based on the question's arithmetic structure alone. By layer 32, Basque math reaches GT rank 62 with 70.3% confidence—reasonable, though still below English (rank 261, 94.2%) and Spanish (rank 452, 81.0%).

Finding 2: Basque factual representations fail to converge. English factual questions reach a mean GT rank of 26 at layer 32—the model is essentially pointing directly at the correct answer token. Spanish factual rank (219) is 8× worse; Basque factual rank (2,440) is 94× worse and effectively random. This catastrophic failure—with the model failing to place the correct answer within the top 2,440 tokens of the distribution—mirrors the 0% behavioral accuracy and confirms that the Basque factual failure is not a generation-time artifact but is present in the model's internal representation. Notably, Basque factual *confidence* remains high (84.8% at L32) despite the catastrophic rank, indicating the model is "decisive but wrong"—it confidently predicts a token, but not the correct one.

Finding 3: Reasoning confidence reveals "undecided" processing. Basque reasoning top-1 confidence at layer 32 is only 53.9%, far below English (87.0%) and Spanish (68.8%). Unlike Basque factual—which is decisive but wrong—Basque reasoning produces diffuse output distributions, consistent with the model failing to arrive at any coherent answer. This matches the 5% behavioral accuracy.

4.3 RSA: Cross-Lingual Representational Alignment

Table 3: RSA Spearman r between language-pair RDMs at selected layers, for Mistral-7B. Bold: highest correlation per row. Underline: negative (anti-correlated) values.

Category	Pair	L1	L8	L16	L24	L32
Math	EN–ES	**0.906**	0.871	0.834	0.858	0.785
	EN–EU	**0.807**	0.683	0.299	0.352	0.389
	ES–EU	**0.846**	0.630	0.496	0.465	0.273
Factual	EN–ES	0.718	**0.840**	0.704	0.795	0.782
	EN–EU	0.508	0.406	−0.032	0.142	−0.049
	ES–EU	0.694	0.384	0.056	0.166	0.013
Reasoning	EN–ES	0.394	**0.810**	0.706	0.739	0.558
	EN–EU	0.303	−0.097	0.088	0.023	0.018
	ES–EU	0.584	−0.035	0.104	0.022	−0.080

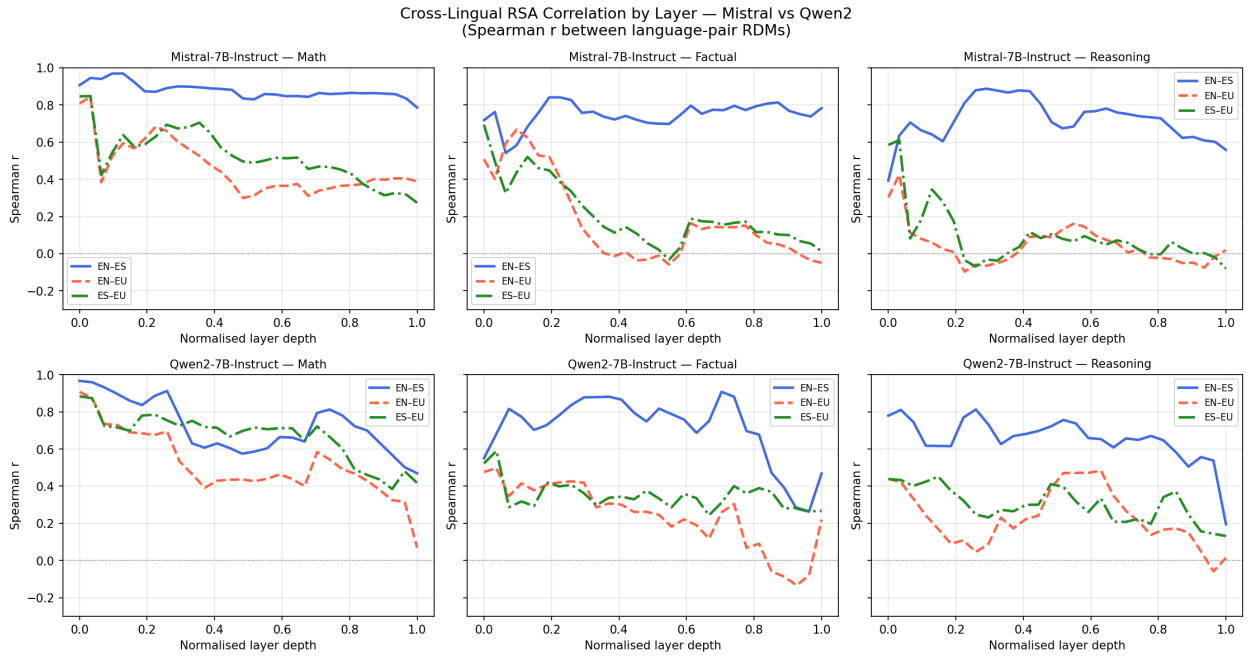

Figure 4: RSA Spearman r curves across all layers, for Mistral-7B (top) and Qwen2-7B (bottom), for each language pair and task category. The mid-layer peak for EN–ES reasoning is clearly visible in Mistral (top center). Qwen2 generally shows lower and noisier RSA values, despite its higher Basque accuracy.

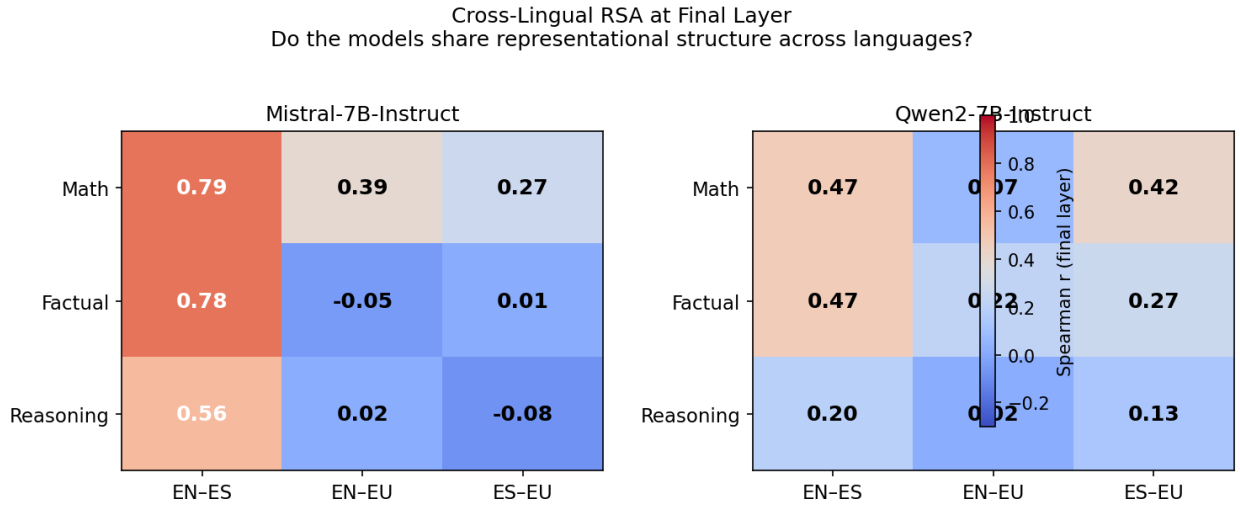

Figure 5: RSA Spearman r at the final layer for all language pairs and categories (Mistral left, Qwen2 right). The EN–EU factual cell is near zero or negative for both models, while EN–ES math maintains the highest alignment.

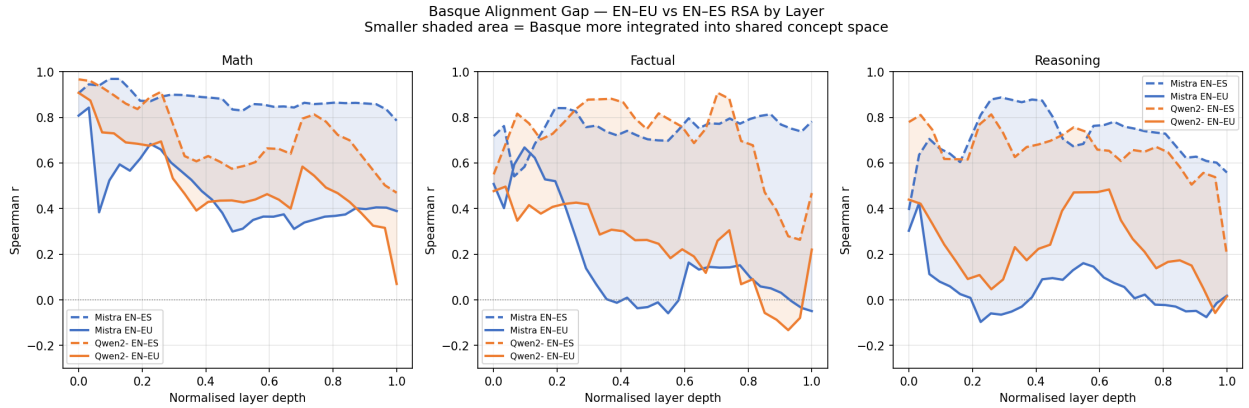

Figure 6: EN–EU vs. EN–ES RSA gap by layer, for each task category (Mistral-7B). Shaded region marks the gap between the two curves. The gap is largest for factual and reasoning, where Basque representations diverge from English while Spanish remains aligned.

Table 3 and Figures 4–6 reveal four findings:

Finding 4: Math has the strongest and most durable cross-lingual alignment. EN–ES math RSA begins at $r = 0.906$ at layer 1 and remains at $r = 0.785$ at layer 32—the highest final-layer correlation across all categories and language pairs. Even EN–EU math maintains a positive correlation (0.389) through all 32 layers, despite Basque's very different surface form. This indicates that the model internally organizes math questions by their mathematical structure (the operations, magnitudes, and expected answer types) rather than by the language they are written in—a form of language-agnostic representational geometry for arithmetic content.

Finding 5: Basque-English factual alignment becomes structurally random. The EN–EU factual RSA drops from 0.508 at layer 1 to -0.049 at layer 32, statistically indistinguishable from zero and occasionally anti-correlated. The relational structure among Basque factual questions has no correspondence to the structure among the same questions in English. We interpret this as the model treating Basque factual content as noise: it tokenizes the input but does not route it through the factual knowledge retrieval circuits it uses for English. The model's representation of "which Basque questions are similar" has no alignment with which English questions are similar.

Finding 6: Reasoning EN–ES peaks in mid-layers—evidence for implicit translation. The EN–ES reasoning RSA rises from 0.394 at layer 1 to a peak of 0.810 at layer 8, then declines to 0.558 at layer 32. This non-monotonic mid-layer peak is not present in math (which is high throughout) or factual (which is high in early-to-mid layers and stays elevated). The peak-then-decline pattern is consistent with an implicit translation mechanism: the model first maps both English and Spanish reasoning problems to a shared internal representation (mid-layers), then resolves the answer in a language-specific way at the output stage (late layers). The mid-layer peak implies the model's reasoning machinery is cross-lingual, even though its output generation is not.

Finding 7: Early layers show language-agnostic structure even for Basque. Despite Basque factual representations collapsing by layer 32, the layer-1 RSA for EN–EU factual (0.508) and EN–EU reasoning (0.303) are moderate—higher than expected for a near-random language. We attribute this to the earliest transformer layers extracting surface-level features (BPE subword overlap, positional encodings, character-level n-grams) that are partially shared across languages before the model's language-specific processing diverges in middle and final layers.

4.4 Cross-Model Comparison

Table 4: RSA Spearman r at the final layer, compared between Mistral-7B and Qwen2-7B. Bold indicates the model with higher RSA per cell. Qwen2 achieves better Basque accuracy but shows *lower* EN–ES RSA, suggesting different internal multilingual representations.

Category	Pair	Mistral	Qwen2	Δ
Math	EN–ES	**0.785**	0.470	−0.316
	EN–EU	**0.389**	0.070	−0.319
	ES–EU	0.273	**0.417**	+0.144
Factual	EN–ES	**0.782**	0.468	−0.314
	EN–EU	−0.049	**0.220**	+0.269
	ES–EU	0.013	**0.268**	+0.255
Reasoning	EN–ES	**0.558**	0.196	−0.362
	EN–EU	0.018	**0.015**	−0.003
	ES–EU	−0.080	**0.133**	+0.212

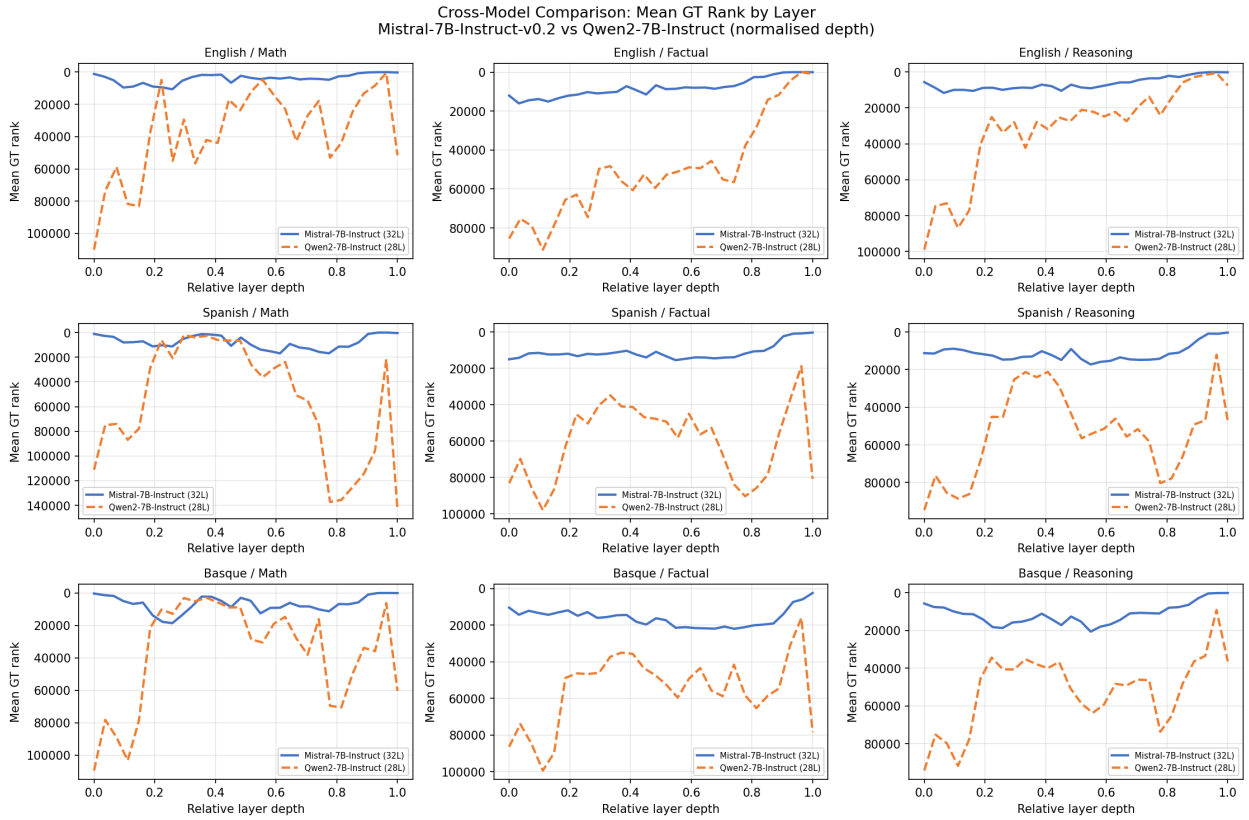

Figure 7: Cross-model GT rank trajectories for each language and category. Mistral-7B (solid) and Qwen2-7B (dashed) are overlaid. Qwen2's higher absolute rank values reflect its larger vocabulary (152K vs. 32K), making direct rank comparison across models uninformative; the trajectory *shapes* are more comparable.

Table 4 reveals a striking dissociation: Qwen2 achieves far better Basque accuracy (37% vs. 8%), yet Mistral shows consistently *higher* EN–ES RSA at the final layer across all categories (e.g., math: 0.785 vs. 0.470; reasoning: 0.558 vs. 0.196). Simultaneously, Qwen2 shows better EN–EU and ES–EU RSA for factual and reasoning, consistent with its greater Basque ability.

This dissociation—better Basque accuracy but less English-Spanish alignment—suggests the two models employ qualitatively different strategies for multilingual processing. Mistral appears to develop strong cross-lingual representational alignment between English and Spanish (two similar Indo-European languages) but has nearly no usable representation for Basque. Qwen2, by contrast, seems to develop a more uniformly multilingual representational space that provides some handle on Basque but does not show as strong English-Spanish convergence. This is consistent with Qwen2's training data including more diverse multilingual content, which may prevent over-specialization toward the English-Spanish axis.

5 Discussion

The Basque wall. Both models show a stark performance discontinuity between Spanish and Basque. For Mistral, this discontinuity is both behavioral (60% $\rightarrow$ 8%) and mechanistic (EN–EU factual RSA 0.782 $\rightarrow -0.049$). Basque is a language isolate: it shares no common ancestor with Indo-European languages, has unique morphological structure, and is almost certainly underrepresented in both models' training corpora. Our results suggest that below a critical training data threshold, a model may tokenize a language without meaningfully processing it—producing confident, wrong answers for factual questions, or diffuse, uncertain outputs for reasoning.

Math as a language-agnostic domain. Across all analyses, math exhibits the weakest language dependence: the best early-layer GT ranks (Basque math, layer 1), the highest EN–EU RSA at the final layer (0.389 for Mistral), and the smallest accuracy gap between English and Spanish (45% vs. 50% for Mistral). We attribute this to the symbolic nature of mathematical answers: the model can identify the correct numeric token from the structure of the problem regardless of which language the problem is posed in, because the answer token itself is language-invariant. This suggests that tasks with language-invariant answer spaces may be more robust to multilingual generalization gaps.

The implicit translation hypothesis. The RSA mid-layer peak for EN–ES reasoning ($r = 0.394$ at L1, $r = 0.810$ at L8, $r = 0.558$ at L32) provides the clearest mechanistic evidence for implicit translation in our data. The interpretation is that the model first constructs distinct per-language representations (early layers), then converges to a shared reasoning representation in middle layers—where the structural problem ("who arrives first?", "which number is odd?") is solved in a language-agnostic way—and finally resolves the answer token in a language-specific output format. The decline from L8 to L32 reflects the output-side language specificity, not a failure of cross-lingual understanding.

This finding raises a natural question: why does the mid-layer peak appear for reasoning but not math? Math does not require this convergence because its answer tokens are already language-agnostic; the model can output the numeric token directly from any language's representation. Factual questions, by contrast, depend on knowledge that is represented in a language-specific way in the training corpus, so the model cannot converge to a shared intermediate representation—and for Basque, there is no intermediate representation to converge to at all.

Model architecture and the RSA paradox. Qwen2's larger vocabulary (152K tokens) implies a fundamentally different token-level representation for the same text compared to Mistral (32K tokens). Basque words that were split into many subwords by Mistral's tokenizer may be represented as single tokens in Qwen2's larger vocabulary, improving its ability to process Basque inputs. This tokenization advantage is a plausible explanation for Qwen2's dramatic Basque accuracy improvement (particularly for math, where the correct answer—a numeral—may benefit from consistent tokenization). However, the RSA results suggest that this improved Basque processing does not come via stronger English-Spanish alignment; rather, Qwen2 may process each language more independently, forgoing the strong EN–ES cross-lingual structure that Mistral develops.

Limitations. Our benchmark is small (20 questions per language-category cell), limiting statistical power. The LLM-as-judge evaluation may introduce systematic biases, particularly for Basque, where the judge may have limited ability to evaluate non-English answers. Basque translations were produced by GPT-4, which may have introduced subtle errors. We evaluate only two model families at the 7B scale; results may differ for larger models or models with explicit multilingual training objectives. Finally, our RSA analysis uses the last-token hidden state, which may not capture all relevant information in autoregressive transformers.

6 Conclusion

We present a joint behavioral and mechanistic analysis of cross-lingual generalization in Mistral-7B and Qwen2-7B across English, Spanish, and Basque. Our results show that cross-lingual performance gaps are not surface-level artifacts but are reflected in the structure of residual stream representations at every layer. Basque factual representations become structurally random by the final layer in Mistral, while math representations maintain positive cross-lingual alignment even for Basque. Mid-layer RSA peaks for English-Spanish reasoning provide evidence for an implicit translation stage in the residual stream. Qwen2 is substantially more robust for Basque but achieves this via different internal representations than Mistral, suggesting that multilingual robustness can arise through different mechanistic routes.

Our work contributes to the growing literature on multilingual interpretability and suggests that layer-wise RSA is a useful tool for characterizing *where* and *how* cross-lingual generalization fails or succeeds. Future directions include activation patching—substituting English or Spanish residual streams into Basque forward passes to test whether improved representations improve Basque accuracy—and scaling to a larger benchmark with statistical power sufficient to isolate individual question difficulty from language-level trends.

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